

AHMED IBRAHIM



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👤 PROFILE

AI Research Specialist with proven success developing a deep learning framework that boosted prediction accuracy by 30% and integrating AI features to increase user engagement by 25%. Expert in machine learning algorithms, natural language processing, and AI model development. Strong communicator and critical thinker skilled in cross-cultural collaboration. Offers midoalfa advanced AI research expertise and a track record of driving impactful, data-driven solutions.

📅 PROFESSIONAL EXPERIENCES

📅 03/2021 – TODAY 📍 MUNICH, GERMANY

Product and Test Engineer (Automotive) Infineon Technologies AG

- Responsible for characterization and validation of MOSFET and IGBT and their drive circuits.
- Responsible for analyzing test results from the backend and front end (wafer level, lot level, and chip level) from Automatic Test Equipment.
- Design test and validation for ICs.
- Performed and maintained electrical test methodologies to ensure product compliance with the data sheet specifications.
- Data analysis for yield enhancement, defect detection for different stages of chip and wafer, and analysis of the effect of deviations on product characteristics.
- Created a cross-domain mapping tool to map results from the test domain to the simulation domain.
- In charge of different project statistical analyses to detect faults and keep track of product flow and deviations.
- Performed design, wafer processing, and simulation design of experiments (DOEs) to determine the optimum process and device performance characteristics.

📅 07/2018 – 09/2019 📍 GURUGRAM, INDIA

Executive Engineer Siemens

📅 PROFESSIONAL EXPERIENCES

- Characterization of semiconductor devices and defect analysis to find out the root causes of failure.
- Design and implementation of different converter topologies.
- Design and implementation of IGBT based Voltage source converter circuit in HVDC.
- Developed a web interface that connected the database with an internal simulation tool for automatic periodic and scheduled reporting of simulation results.
- Designed and developed switch-mode power supplies and their integration into existing power supply systems.

📅 06/2016 – 07/2017 📍 DELHI, INDIA

Intern Siemens

- Process Instrumentation.
- AC/DC drives.
- Industrial automation.

📅 09/2020 – 10/2021 📍 MUNICH, GERMANY

Driver Design Engineer (Volunteer) TUM Hyperloop

- Responsible for driver design for MOSFET based converter.
- Calculation and product selection for dc-link capacitor for MOSFET based full-bridge converter.
- Designed pre-charging and discharging circuits for DC link capacitor.

🔧 PROJECTS

📅 07/2019

Over/Under Voltage Tripping Circuit for Distributed System Load with GSM Alert Using Microcontroller Jamia Millia Islamia

- Published research paper in International Journal of Innovative Technology and Exploring Engineering (IJITEE).
- Using mobile phone to control and monitor



PROJECTS

electrical circuits by sending text messages and getting message alerts during overload.

10/2018

Design and Simulation of DC Voltage Multiplier Using 555 Timer Jamia Millia Islamia

- The research project focused on the development of 4 stages of the cascaded multiplier, each stage adding 12 volts, stepping up 12V DC input to 48V DC output.
- Published article in International Journal of Advanced Research, Ideas and Innovations in Technology.



HOBBIES



Exploring
distant lands



Books



Football



EDUCATIONS

10/2019 – 06/2021 MUNICH, GERMANY

Master of Science in Power Engineering

Major: Circuit Design for Automotive MOSFET

03/2014 – 06/2018 DELHI, INDIA

Electrical Engineering Jamia Millia Islamia

Specialization in Power Electronics and Converter Design

SKILLS

LANGUAGES

English

German

TOOLS AND SOFTWARE

MATLAB Simulink

Python

Altium

Labview

PCB design

C++

MS Office



STRENGTHS

Adaptable # Responsible # Self-motivated
Strong work ethic # Creative thinking
Honesty